Exercise 10.4 SOLUTION

Q1. Express the following products as sum or differences.

i. $2\sin 3\theta \cos\theta$

Sol: $= \sin(3\theta + \theta) + \sin(3\theta - \theta)$

$= \sin 4\theta + \sin 2\theta$

ii. $2\cos 5\theta \sin 3\theta$

Sol: $= \sin(5\theta + 3\theta) - \sin(5\theta - 3\theta)$

$= \sin 8\theta - \sin 2\theta$

iii. $\sin 5\theta \cos 2\theta$

Sol: $= \frac{1}{2}(2\sin 5\theta \cos 2\theta) = \frac{1}{2}[\sin(5\theta + 2\theta) + \sin(5\theta - 2\theta)]$

$= \frac{1}{2}(\sin 7\theta + \sin 3\theta)$

iv. $2\sin 7\theta \sin 2\theta$

Sol: $= -(-2\sin 7\theta \sin 2\theta)$

$= -[\cos(7\theta + 2\theta) - \cos(7\theta - 2\theta)]$

$= -(\cos 9\theta - \cos 5\theta)$

$= \cos 5\theta - \cos 9\theta$

v. $\cos(x + y)\sin(x - y)$

Sol: $= \frac{1}{2}\left(2\cos(x+y)\sin(x-y)\right) = \frac{1}{2}\left(\sin((x+y)+(x-y)) - \sin(x+y) - (x-y)\right)$

$= \frac{1}{2}\left(\sin(x + \cancel{y} + x - \cancel{y}) - \sin(\cancel{x} + y - \cancel{x} + y)\right) = \frac{1}{2}(\sin 2x - \sin 2y)$

vi. $\cos(2x + 30°)\cos(2x - 30°)$

Sol. $= \frac{1}{2}\left(2\cos(2x + 30^\circ)\cos(2x - 30^\circ)\right) = \frac{1}{2}\left[\cos(2x + 30^\circ + 2x - 30^\circ) + \cos((2x + 30^\circ) - (2x - 30^\circ))\right]$

$= \frac{1}{2}\left[\cos(2x + \cancel{30^\circ} + 2x - \cancel{30^\circ}) + \cos(\cancel{2x} + 30^\circ - \cancel{2x} + 30^\circ)\right] = \frac{1}{2}(\cos 4x + \cos 60^\circ)$

vii. $\sin 12° \sin 46°$

Sol. $= \frac{-1}{2}(-2\sin 12^\circ \sin 46^\circ) = -\frac{1}{2}(\cos(12^\circ + 46^\circ) - \cos(12^\circ - 46^\circ))$

$= -\frac{1}{2}(\cos 58^\circ - \cos(-34^\circ)) = -\frac{1}{2}(\cos 58^\circ - \cos 34^\circ) = \frac{1}{2}(\cos 34^\circ - \cos 58^\circ)$

viii. $\sin(x + 45°)\sin(x - 45°)$

Sol. $= -\frac{1}{2}\left(-2\sin(x + 45^\circ)\sin(x - 45^\circ)\right) = -\frac{1}{2}\left[\cos\left[(x+45^\circ)+(x-45^\circ)\right] - \cos\left[(x+45^\circ)-(x-45^\circ)\right]\right]$

$= -\frac{1}{2}\left[\cos(x + \cancel{45^\circ} + x - \cancel{45^\circ}) - \cos(\cancel{x} + 45^\circ - \cancel{x} + 45^\circ)\right]$

$= -\frac{1}{2}\left[\cos 2x - \cos 90^\circ\right] = \frac{1}{2}(-\cos 2x + \cos 90^\circ) = \frac{1}{2}(\cos 90^\circ - \cos 2x)$

Q2. Express the following sums or differences as products

i $\sin 5\theta + \sin 3\theta$

Sol: $= 2\sin\frac{5\theta + 3\theta}{2}\cos\frac{5\theta - 3\theta}{2} = 2\sin 4\theta \cos\theta$

ii. $\sin 8\theta - \sin 4\theta$

Sol. $= 2\cos\frac{8\theta + 4\theta}{2}\sin\frac{8\theta - 4\theta}{2} = 2\cos 6\theta \sin 2\theta$

iii. $\cos 6\theta + \cos 3\theta$

Sol. $= 2\cos\frac{6\theta + 3\theta}{2}\cos\frac{6\theta - 3\theta}{2} = 2\cos\frac{9\theta}{2}\cos\frac{3\theta}{2}$

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iv. $\text{Cos}7\theta - \text{Cos}\theta$

Sol. $= -2\text{Sin}\dfrac{7\theta+\theta}{2}\,Sin\dfrac{7\theta-\theta}{2} = -2\,\text{Sin}\,4\theta\,\text{Sin}3\theta$

v. $\text{Cos}12^\circ + \text{Cos}48^\circ$

Sol. $= 2\text{Cos}\dfrac{12^\circ+48^\circ}{2}\,Cos\dfrac{12^\circ-48^\circ}{2}$

$= 2\text{Cos}\,\dfrac{60^\circ}{2}\,\text{Cos}\,\dfrac{(-36^\circ)}{2}$ $\qquad \boxed{\text{Cos}(-\theta) = \text{Cos}\theta}$

$= 2\text{Cos}30^\circ\text{Cos}(18^\circ)$

vi. $\text{Sin}\,(x + 30^\circ) + \text{Sin}\,(x - 30^\circ)$

Sol. $= 2\text{Sin}\left(\dfrac{x+\cancel{30^\circ}+x-\cancel{30^\circ}}{2}\right)\text{Cos}\,\dfrac{(x+30^\circ)-(x-30^\circ)}{2}$

$= 2\text{Sin}\left(\dfrac{\cancel{2}x}{\cancel{2}}\right)\text{Cos}\left(\dfrac{\cancel{x}+30^\circ-\cancel{x}+30^\circ}{2}\right) = 2\text{Sin}x\text{Cos}30^\circ$

Q3. Prove the following identities

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i $\dfrac{Sin3x-Sinx}{Cosx-Cos3x} = \text{Cot}2x$

Sol. L.H.S $= \dfrac{Sin3x-Sinx}{Cosx-Cos3x}$

$= \dfrac{2Cos\dfrac{3x+x}{2}Sin\dfrac{3x-x}{2}}{-2Sin\dfrac{3x+x}{2}Sin\dfrac{x-3x}{2}} = \dfrac{Cos2xSinx}{-Sin2xSin(-x)} = \dfrac{Cos2xSinx}{-Sin2x(-Sinx)} = \dfrac{Cos2x\,\cancel{Sinx}}{Sin2x\,\cancel{Sinx}}$

$= \text{Cot}2x = \text{R.H.S}$

ii. $\dfrac{Sin8x+Sin2x}{Cos8x+Cos2x} = \text{tan}5x$

Sol. L.H.S $= \dfrac{Sin8x+Sin2x}{Cos8x+Cos2x} = \dfrac{2Sin\dfrac{8x+2x}{2}Cos\dfrac{8x-2x}{2}}{2Cos\dfrac{8x+2x}{2}Cos\dfrac{8x-2x}{2}}$

$= \dfrac{Sin5x\,\cancel{Cos3x}}{Cos5x\,\cancel{Cos3x}} = \text{tan}5x = \text{R.H.S}$

iii. $\dfrac{\text{Sin}\,A - \text{Sin}\,B}{\text{Sin}\,A + \text{Sin}\,B} = \text{tan}\,\dfrac{A-B}{2}\,\text{Cot}\,\dfrac{A+B}{2}$

Sol. L.H.S $= \dfrac{\text{Sin}\,A - \text{Sin}\,B}{\text{Sin}\,A + \text{Sin}\,B}$

$= \dfrac{\cancel{2}Cos(\frac{A+B}{2})Sin(\frac{A-B}{2})}{\cancel{2}Sin(\frac{A+B}{2})Cos(\frac{A-B}{2})} = \dfrac{\text{Cos}\left(\frac{A+B}{2}\right)}{\text{Sin}\left(\frac{A+B}{2}\right)} \cdot \dfrac{\text{Sin}\left(\frac{A-B}{2}\right)}{\text{Cos}\left(\frac{A-B}{2}\right)}$

$= \text{Cot}\left(\dfrac{A+B}{2}\right)\text{tan}\left(\dfrac{A-B}{2}\right) = \text{tan}\left(\dfrac{A-B}{2}\right)\text{Cot}\left(\dfrac{A+B}{2}\right)$

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iv. $\dfrac{\sin 80^0 + \sin 40^0}{\cos 80^0 + \cos 40^0} = \sqrt{3}$

Sol: $L.H.S = \dfrac{\sin 80^0 + \sin 40^0}{\cos 80^0 + \cos 40^0}$

$$= \frac{2\sin\left(\frac{80^0+40^0}{2}\right)\cos\left(\frac{80^0-40^0}{2}\right)}{2\cos\left(\frac{80^0+40^0}{2}\right)\cos\left(\frac{80^0-40^0}{2}\right)} = \frac{\sin 60^0 \cos 20^0}{\cos 60^0 \cos 20^0} = \frac{\sin 60^0}{\cos 60^0} = \tan 60^0 = \sqrt{3} = R.H.S$$

Q4. Prove that:

Note: Wrong statement is given. It cannot be solved with $\cos 160^0$

i. $\cos 15^0 + \cos 105^0 + \cos 195^0 + \cos 285^0 = 0$

Sol:

L.H.S $\cos 15^0 + \cos 105^0 + \cos 195^0 + \cos 285^0$

$= \cos 15^0 + \cos(90^0 + 15^0) + \cos(180^0 + 15^0) + \cos(270^0 + 15^0) = \cos 15^0 - \sin 15^0 - \cos 15^0 + \sin 15^0$

$= 0 = R.H.S$

ii. $\dfrac{\sin 2\theta + \sin 4\theta + \sin 6\theta + \sin 8\theta}{\cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta} = \tan 5\theta$

Sol: $L.H.S = \dfrac{\sin 2\theta + \sin 4\theta + \sin 6\theta + \sin 8\theta}{\cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta}$

$$= \frac{2\sin\left(\frac{2\theta+4\theta}{2}\right)\cos\left(\frac{2\theta-4\theta}{2}\right) + 2\sin\left(\frac{6\theta+8\theta}{2}\right)\cos\left(\frac{6\theta-8\theta}{2}\right)}{2\cos\left(\frac{2\theta+4\theta}{2}\right)\cos\left(\frac{2\theta-4\theta}{2}\right) + 2\cos\left(\frac{6\theta+8\theta}{2}\right)\cos\left(\frac{6\theta-8\theta}{2}\right)}$$

$$= \frac{2\sin\left(\frac{6\theta}{2}\right)\cos\left(\frac{-2\theta}{2}\right) + 2\sin\left(\frac{14\theta}{2}\right)\cos\left(\frac{-2\theta}{2}\right)}{2\cos\left(\frac{6\theta}{2}\right)\cos\left(\frac{-2\theta}{2}\right) + 2\cos\left(\frac{14\theta}{2}\right)\cos\left(\frac{-2\theta}{2}\right)} = \frac{2\sin 3\theta\cos\theta + 2\sin 7\theta\cos\theta}{2\cos 3\theta\cos\theta + 2\cos 7\theta\cos\theta}$$

$$= \frac{2\cos\theta\,(\sin 3\theta + \sin 7\theta)}{2\cos\theta\,[\cos 3\theta + \cos 7\theta]} = \frac{2\sin 5\theta\cos(-2\theta)}{2\cos 5\theta\cos(-2\theta)} = \frac{\sin 5\theta}{\cos 5\theta} = \tan 5\theta = R.H.S$$

iii. $\cos^2\left(\frac{\pi}{4} - \frac{\alpha}{2}\right) - \cos^2\left(\frac{\pi}{4} + \frac{\alpha}{2}\right) = \sin\alpha$

Sol: $L.H.S = \cos^2\left(\frac{\pi}{4} - \frac{\alpha}{2}\right) - \cos^2\left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$

$$= \left[\cos\left(\frac{\pi}{4} - \frac{\alpha}{2}\right) - \cos\left(\frac{\pi}{4} + \frac{\alpha}{2}\right)\right]\left[\cos\left(\frac{\pi}{4} - \frac{\alpha}{2}\right) + \cos\left(\frac{\pi}{4} + \frac{\alpha}{2}\right)\right]$$

$$= \left[-2\sin\left(\frac{\frac{\pi}{4} - \frac{\alpha}{2} + \frac{\pi}{4} + \frac{\alpha}{2}}{2}\right)\sin\left(\frac{\frac{\pi}{4} - \frac{\alpha}{2} - \frac{\pi}{4} - \frac{\alpha}{2}}{2}\right)\right]\left[2\cos\left(\frac{\frac{\pi}{4} - \frac{\alpha}{2} + \frac{\pi}{4} + \frac{\alpha}{2}}{2}\right)\cos\left(\frac{\frac{\pi}{4} - \frac{\alpha}{2} - \frac{\pi}{4} - \frac{\alpha}{2}}{2}\right)\right]$$

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$$= \left[-2\sin\left(\frac{\frac{2\pi}{4}}{2}\right)\sin\left(\frac{\frac{-2\alpha}{2}}{2}\right)\right]\left[2\cos\left(\frac{\frac{2\pi}{4}}{2}\right)\cos\left(\frac{\frac{-2\alpha}{2}}{2}\right)\right] = \left[-2\sin\left(\frac{\pi}{4}\right)\sin\left(\frac{-\alpha}{2}\right)\right]\left[2\cos\left(\frac{\pi}{4}\right)\cos\left(-\frac{\alpha}{2}\right)\right]$$

$$= \left[2.\frac{1}{\sqrt{2}}.\sin\frac{\alpha}{2}\right]\left[2.\frac{1}{\sqrt{2}}.\cos\frac{\alpha}{2}\right] = \frac{4}{2}\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}$$

$$= 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2} = \sin\alpha = R.H.S$$

iv. $\text{Sin}\left(\frac{\pi}{4}-\theta\right)\text{Sin}\left(\frac{\pi}{4}+\theta\right) = \frac{1}{2}\text{Cos}2\theta$

Sol: $\text{L.H.S} = \text{Sin}\left(\frac{\pi}{4}-\theta\right)\text{Sin}\left(\frac{\pi}{4}+\theta\right)$

$$= -\frac{1}{2}\left[-2\sin\left(\frac{\pi}{4}-\theta\right)\text{Sin}\left(\frac{\pi}{4}+\theta\right)\right]$$

$$= -\frac{1}{2}\left[\cos\left(\frac{\pi}{4}-\cancel{\theta}+\frac{\pi}{4}+\cancel{\theta}\right)-\text{Cos}\left[\left(\frac{\pi}{4}-\theta\right)-\left(\frac{\pi}{4}+\theta\right)\right]\right]$$

$$= -\frac{1}{2}\left[\cos\left(2\times\frac{\pi}{4}\right)-\text{Cos}\left(\cancel{\frac{\pi}{4}}-\theta-\cancel{\frac{\pi}{4}}-\theta\right)\right]$$

$$= -\frac{1}{2}\left[\text{Cos}\frac{\pi}{2}-\text{Cos}(-2\theta)\right]$$

$$= -\frac{1}{2}[0-\text{Cos}2\theta] = \frac{1}{2}\text{Cos}2\theta = \text{R.H.S}$$

v. $$\frac{Sin\theta+Sin3\theta+Sin5\theta+Sin7\theta}{Cos\theta+Cos3\theta+Cos5\theta+Cos7\theta} = \tan 4\theta$$

Sol: $$\text{L.H.S} = \frac{Sin\theta+Sin3\theta+Sin5\theta+Sin7\theta}{Cos\theta+Cos3\theta+Cos5\theta+Cos7\theta} = \frac{Sin7\theta+Sin\theta+Sin5\theta+Sin3\theta}{Cos7\theta+Cos\theta+Cos5\theta+Cos3\theta}$$

$$= \frac{2Sin\frac{7\theta+\theta}{2}Cos\frac{7\theta-\theta}{2}+2Sin\frac{5\theta+3\theta}{2}Cos\frac{5\theta-3\theta}{2}}{2Cos\frac{7\theta+\theta}{2}Cos\frac{7\theta-\theta}{2}+2Cos\frac{5\theta+3\theta}{2}Cos\frac{5\theta-3\theta}{2}}$$

$$= \frac{2Sin4\theta.Cos3\theta+2Sin4\theta.Cos\theta}{2Cos4\theta.Cos3\theta+2Cos4\theta.Cos\theta} = \frac{\cancel{2}Sin4\theta\cancel{[Cos3\theta+Cos\theta]}}{\cancel{2}Cos4\theta\cancel{[Cos3\theta+Cos\theta]}} = \frac{Sin4\theta}{Cos4\theta}$$

$$= \tan 4\theta = R.H.S$$

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5. Prove that

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i $\text{Cos}20^\circ\,\text{Cos}40^\circ\,\text{Cos}60^\circ\,\text{Cos}80^\circ = \frac{1}{16}$

Sol: $\text{L.H.S} = \text{Cos}20^\circ\,\text{Cos}40^\circ\,\text{Cos}60^\circ\,\text{Cos}80^\circ$

$$= \text{Cos}20^\circ\,\text{Cos}40^\circ\left(\frac{1}{2}\right)\text{Cos}80^\circ$$

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$= \frac{1}{2}$ (Cos20° Cos40°) Cos80°

$= \frac{1}{4}$ (2Cos20° Cos40°) Cos80° ('x' & '÷' by 2)

$= \frac{1}{4}$ (Cos (20° + 40°)+ Cos(20° –40°)] Cos80°

$= \frac{1}{4}$ [(Cos (20° + 40°) Cos80° +(Cos (20° – 40°)) Cos80°]

$= \frac{1}{4}$ (Cos60° Cos80° + Cos20° Cos80°) (Cos(–20°)= Cos20°)

$= = \frac{1}{4}\left(\frac{1}{2}Cos80'' + Cos20''Cos80''\right) = \frac{1}{4}\left(\frac{Cos80'' + 2Cos20''Cos80''}{2}\right)$

$= \frac{1}{8}$ (Cos80° + Cos (20° + 80°) + Cos (20° – 80°))

$= \frac{1}{8}$ (Cos(180° – 100°) + Cos (100°) + Cos (– 60°)]

$= \frac{1}{8}$ [Cos180° Cos 100° + Sins180° Sin100° + Cos100° + Cos 60°]

$= \frac{1}{8}$ [(-1) Cos 100° + (0) Sin100° + Cos100° + $\frac{1}{2}$]

$= \frac{1}{8}\left[-\cancel{Cos\,100^\circ} + \cancel{Cos100^\circ} + \frac{1}{2}\right]$

$= \frac{1}{8}\left(\frac{1}{2}\right) = \frac{1}{16} = R.H.S$

ii. $Sin\frac{\pi}{9}Sin\frac{2\pi}{9}Sin\frac{\pi}{3}Sin\frac{4\pi}{9} = \frac{3}{16}$

Sol: L.H.S $= Sin\frac{\pi}{9}Sin\frac{2\pi}{9}Sin\frac{\pi}{3}Sin\frac{4\pi}{9}$

= Sin20°Sin40°Sin60°Sin80°

= Sin20°Sin40° $\frac{\sqrt{3}}{2}$ Sin80°

$= \frac{\sqrt{3}}{2}$ (Sin20°Sin40°)Sin80° = $-\frac{\sqrt{3}}{4}$ (–2 Sin20°Sin40°)Sin80°

$= -\frac{\sqrt{3}}{4}$ [Cos (20° +40°) – Cos (20° – 40°)] Sin 80°

$= \frac{-\sqrt{3}}{4}$ [Cos 60° – Cos (– 20°)] Sin80° $= \frac{-\sqrt{3}}{4}\left(\frac{1}{2} - Cos20^\circ\right)Sin80^\circ$

$= -\frac{\sqrt{3}}{4}$ [$\frac{1}{2}$ Sin80° – Cos20°Sin80°]

$= -\frac{\sqrt{3}}{4}$ [$\frac{Sin80'' - 2Cos20''\,Sin80''}{2}$]

$$= -\frac{\sqrt{3}}{8}\{\sin(180^\circ - 100^\circ) - [\sin(20^\circ + 80^\circ) - \sin(20^\circ - 80^\circ)]\}$$

$$= -\frac{\sqrt{3}}{8}\left[\cancel{\sin 100^\circ} - \cancel{\sin 100^\circ} + \sin(-60^\circ)\right]$$

$$= -\frac{\sqrt{3}}{8}(-\sin 60^\circ) = -\frac{\sqrt{3}}{8}\left(-\frac{\sqrt{3}}{2}\right) = \frac{3}{16} = R.H.S$$

iii. $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ = \frac{1}{16}$

Sol. L.H.S $= \sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ$

$$= \sin 10^\circ\left(\frac{1}{2}\right)\sin 50^\circ \sin 70^\circ = \frac{1}{2}(\sin 10^\circ \sin 50^\circ)\sin 70^\circ$$

$$= -\frac{1}{4}(-2\sin 10^\circ \sin 50^\circ)\sin 70^\circ = \frac{1}{4}[\cos(10^\circ + 50^\circ) - \cos(10^\circ - 50^\circ)]\sin 70^\circ$$

$$= -\frac{1}{4}[\cos 60^\circ - \cos(-40^\circ)]\sin 70^\circ = -\frac{1}{4}[\cos 60^\circ \sin 70^\circ - \cos 40^\circ \sin 70^\circ]$$

$$= -\frac{1}{4}\left[\frac{1}{2}\sin 70^\circ - \cos 40^\circ \sin 70^\circ\right] = -\frac{1}{4}\left(\frac{Sin70'' - 2Cos40'' Sin70''}{2}\right)$$

$$= -\frac{1}{8}[\sin(70^\circ) - \{(\sin(40^\circ + 70^\circ) - \sin(40^\circ - 70^\circ))\}] = -\frac{1}{8}[\sin(180^\circ - 110^\circ) - \sin 110^\circ + \sin(-30^\circ)]$$

$$= -\frac{1}{8}\left[\cancel{\sin 110^\circ} - \cancel{\sin 110^\circ} - \sin 30^\circ\right] = -\frac{1}{8}\left[-\frac{1}{2}\right] = \frac{1}{16} = R.H.S$$

Q6. Prove that $\frac{\sin 3\theta}{1+2\cos 2\theta} = \sin\theta$, and deduce the value of $\sin 15^0$.

Sol:

L.H.S $= \frac{\sin 3\theta}{1+2\cos 2\theta} = \frac{3\sin\theta - 4\sin^3\theta}{1+2(1-2\sin^2\theta)} = \frac{\sin\theta(3-4\sin^2\theta)}{1+2-4\sin^2\theta}$

$$= \frac{\sin\theta\cancel{(3-4\sin^2\theta)}}{\cancel{(3-4\sin^2\theta)}} = \sin\theta = R.H.S$$

To deduce $\sin 15^0$

$$\sin 15^0 = \frac{\sin(3\times 15^0)}{1+2\cos(2\times 15^0)} \quad (put\ \theta = 15^\circ\ in\ given\ question)$$

$$= \frac{\sin(45^0)}{1+2\cos 30^0} = \frac{\frac{1}{\sqrt{2}}}{1+2.\frac{\sqrt{3}}{2}} = \frac{\frac{1}{\sqrt{2}}}{1+2.\frac{\sqrt{3}}{2}} = \frac{\frac{1}{\sqrt{2}}}{1+\sqrt{3}} = \frac{1}{\sqrt{2}+\sqrt{6}} = \frac{\sqrt{2}-\sqrt{6}}{(\sqrt{2}+\sqrt{6})(\sqrt{2}-\sqrt{6})}$$

$$= \frac{\sqrt{2}-\sqrt{6}}{(\sqrt{2})^2-(\sqrt{6})^2} = \frac{\sqrt{2}-\sqrt{6}}{2-6} = \frac{\sqrt{2}-\sqrt{6}}{-4} = \frac{\sqrt{6}-\sqrt{2}}{4}$$

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Q7. Prove that $\tan 75^0 - \tan 15^0 = 2\sqrt{3}$

Sol: L.H.S $= \tan 75^0 - \tan 15^0 = \tan\left(45^0 + 30^0\right) - \tan\left(45^0 - 30^0\right)$

$$= \frac{\tan 45^0 + \tan 30^0}{1 - \tan 45^0 \tan 30^0} - \frac{\tan 45^0 - \tan 30^0}{1 + \tan 45^0 \tan 30^0} = \frac{1 + \tan 30^0}{1 - \tan 30^0} - \frac{1 - \tan 30^0}{1 + \tan 30^0}$$

$$= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} - \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{\frac{\sqrt{3}+1}{\sqrt{3}}}{\frac{\sqrt{3}-1}{\sqrt{3}}} - \frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}} = \frac{\sqrt{3}+1}{\cancel{\sqrt{3}}} \times \frac{\cancel{\sqrt{3}}}{\sqrt{3}-1} - \frac{\sqrt{3}-1}{\cancel{\sqrt{3}}} \times \frac{\cancel{\sqrt{3}}}{\sqrt{3}+1}$$

$$= \frac{\sqrt{3}+1}{\sqrt{3}-1} - \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{\left(\sqrt{3}+1\right)^2 - \left(\sqrt{3}-1\right)^2}{\left(\sqrt{3}-1\right)\left(\sqrt{3}+1\right)} = \frac{\left(3+1+2\sqrt{3}\right) - \left(3+1-2\sqrt{3}\right)}{\left(\sqrt{3}\right)^2 - (1)^2}$$

$$= \frac{\cancel{4} + 2\sqrt{3} - \cancel{4} + 2\sqrt{3}}{3-1} = \frac{4\sqrt{3}}{2} = 2\sqrt{3} = R.H.S$$

Q8. Prove that $\cos 15^0 - \sin 15^0 = \frac{1}{\sqrt{2}}$

Sol: L.H.S $= \cos 15^0 - \sin 15^0 = \cos\left(45^0 - 30^0\right) - \sin\left(45^0 - 30^0\right)$

$$= \left[\cos 45^0 \cos 30^0 + \sin 45^0 \sin 30^0\right] - \left[\sin 45^0 \cos 30^0 - \cos 45^0 \sin 30^0\right]$$

$$= \left[\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}\right] - \left[\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2}\right] = \frac{\cancel{\sqrt{3}}}{\cancel{2}\sqrt{2}} + \frac{1}{2\sqrt{2}} - \frac{\cancel{\sqrt{3}}}{\cancel{2}\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} = R.H.S$$

Q9. Prove that $\frac{\sin^2\alpha - \sin^2\beta}{\sin\alpha\cos\alpha - \sin\beta\cos\beta} = \tan(\alpha + \beta)$

Sol: L.H.S $= \frac{\sin^2\alpha - \sin^2\beta}{\sin\alpha\cos\alpha - \sin\beta\cos\beta}$

$$= \frac{(\sin\alpha + \sin\beta)(\sin\alpha - \sin\beta)}{\frac{1}{2}(2\sin\alpha\cos\alpha - 2\sin\beta\cos\beta)} = \frac{2\left[2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}\right]\left[2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}\right]}{\sin 2\alpha - \sin 2\beta}$$

Rearrange numerator

$$= \frac{\cancel{2}\left[2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}\right]\left[2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}\right]}{\cancel{2}\cos\frac{2\alpha+2\beta}{2}\sin\frac{2\alpha-2\beta}{2}} = \frac{\sin\cancel{2}\frac{\alpha+\beta}{\cancel{2}}\sin\cancel{2}\frac{\alpha-\beta}{\cancel{2}}}{\cos(\alpha+\beta).\sin(\alpha-\beta)} = \frac{\sin(\alpha+\beta).\cancel{\sin(\alpha-\beta)}}{\cos(\alpha+\beta).\cancel{\sin(\alpha-\beta)}}$$

$$\tan(\alpha+\beta) = R.H.S$$

2nd Method

$$R.H.S = \tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{\frac{\sin\alpha}{\cos\alpha} + \frac{\sin\beta}{\cos\beta}}{1 - \frac{\sin\alpha}{\cos\alpha}\frac{\sin\beta}{\cos\beta}}$$

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$$= \frac{\dfrac{\sin\alpha\cos\beta+\cos\alpha\sin\beta}{\cos\alpha\cos\beta}}{\dfrac{\cos\alpha\cos\beta-\sin\alpha\sin\beta}{\cos\alpha\cos\beta}} = \frac{\sin\alpha\cos\beta+\cos\alpha\sin\beta}{\cos\alpha\cos\beta-\sin\alpha\sin\beta}\times\frac{\sin\alpha\cos\beta-\cos\alpha\sin\beta}{\sin\alpha\cos\beta-\cos\alpha\sin\beta}$$

$$= \frac{\sin^2\alpha\cos^2\beta-\cos^2\alpha\sin^2\beta}{\cos\alpha\cos\beta\sin\alpha\cos\beta-\cos\alpha\cos\beta\cos\alpha\sin\beta-\sin\alpha\sin\beta\sin\alpha\cos\beta+\sin\alpha\sin\beta\cos\alpha\sin\beta}$$

$$= \frac{\sin^2\alpha\left(1-\sin^2\beta\right)-\left(1-\sin^2\alpha\right)\sin^2\beta}{\sin\alpha\cos\alpha\cos^2\beta-\cos^2\alpha\cos\beta\sin\beta-\sin^2\alpha\sin\beta\cos\beta+\sin\alpha\cos\alpha\sin^2\beta}$$

$$= \frac{\sin^2\alpha-\cancel{\sin^2\alpha\sin^2\beta}-\sin^2\beta+\cancel{\sin^2\alpha\sin^2\beta}}{\sin\alpha\cos\alpha\left(1-\sin^2\beta\right)-\left(1-\sin^2\alpha\right)\cos\beta\sin\beta-\sin^2\alpha\sin\beta\cos\beta+\sin\alpha\cos\alpha\sin^2\beta}$$

$$= \frac{\sin^2\alpha-\sin^2\beta}{\sin\alpha\cos\alpha-\cancel{\sin\alpha\cos\alpha\sin^2\beta}-\cos\beta\sin\beta+\cancel{\sin^2\alpha\sin\beta\cos\beta}-\cancel{\sin^2\alpha\sin\beta\cos\beta}+\cancel{\sin^2\beta\sin\alpha\cos\alpha}}$$

$$= \frac{\sin^2\alpha-\sin^2\beta}{\sin\alpha\cos\alpha-\sin\beta\cos\beta} = L.H.S$$

Q10. **Prove that** $\sin\alpha+\sin\beta+\sin\gamma-\sin(\alpha+\beta+\gamma)=4\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\beta+\gamma}{2}\right)\sin\left(\frac{\gamma+\alpha}{2}\right)$

Sol: L.H.S $=\sin\alpha+\sin\beta+\sin\gamma-\sin(\alpha+\beta+\gamma)$

$$=(\sin\alpha+\sin\beta)+(\sin\gamma-\sin(\alpha+\beta+\gamma))$$

$$=2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)+2\cos\left(\frac{\gamma+\alpha+\beta+\gamma}{2}\right)\sin\left(\frac{\gamma-\alpha-\beta-\gamma}{2}\right)$$

$$=2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)+2\cos\left(\frac{\alpha+\beta+2\gamma}{2}\right)\left(-\sin\left(\frac{\alpha+\beta}{2}\right)\right)$$

$$=2\sin\left(\frac{\alpha+\beta}{2}\right)\left[\cos\left(\frac{\alpha-\beta}{2}\right)-\cos\left(\frac{\alpha+\beta+2\gamma}{2}\right)\right]$$

$$=2\sin\left(\frac{\alpha+\beta}{2}\right)\left\{-2\sin\frac{1}{2}\left(\frac{\alpha-\beta}{2}+\frac{\alpha+\beta+2\gamma}{2}\right)\sin\frac{1}{2}\left(\frac{\alpha-\beta}{2}-\frac{\alpha+\beta+2\gamma}{2}\right)\right\}$$

$$=-4\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta+\alpha+\beta+2\gamma}{2\times2}\right)\sin\left(\frac{\alpha-\beta-\alpha-\beta-2\gamma}{2\times2}\right)$$

$$=-4\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{2\alpha+2\gamma}{2\times2}\right)\sin\left(\frac{-2\beta-2\gamma}{2\times2}\right)$$

$$=-4\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha+\gamma}{2}\right)\sin\left(\frac{-(\beta+\gamma)}{2}\right)$$

$$=4\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\beta+\gamma}{2}\right)\sin\left(\frac{\gamma+\alpha}{2}\right)=R.H.S$$

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